

Project Initialization and Planning Phase

Project Proposal (Proposed Solution) report

Date	2 October 2026
Team ID	SWTID-2026-8368
Project Title	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

The proposal report aims to make illustrated storytelling effortless using generative AI, so that anyone can turn an idea into a finished comic. It tackles the skill and tooling gap between writing a story, drawing it and laying it out, promising faster creation and a shareable PDF. Key features include a Gemini-based outline and story writer and locally generated Stable Diffusion panel images.

Project Overview

Objective	The primary objective is to turn a one-line story idea into a five-panel illustrated comic with captions, narration and a PDF export, with no drawing skills required.
Scope	A local web application: input form, Gemini outline and story generation, Stable Diffusion image generation for five panels, layout builder, preview page, PDF export, JSON API and tests.

Problem Statement

Description	Making a comic needs story writing, consistent illustration and layout. Most people have ideas but lack the art and layout skills, and existing tools give single, disconnected images.
Impact	Solving this lets students, teachers and creators produce engaging visual stories in minutes, improving learning, sharing and creativity.

Proposed Solution

Approach	A guided pipeline: Gemini Flash creates a 5-panel JSON outline; a second Gemini model writes captions and narration while Stable Diffusion v1.5 draws each panel in parallel; a layout builder merges them; fpdf2 exports the PDF.
Key Features	- Setting, tone and art-style pickers - Structured JSON outline - Parallel text and image generation with a warmed-up, cached image model - Preview page and PDF export with Unicode fonts - JSON API and /test-image endpoint

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware	Computing Resources	Apple M5 (MPS) used; any CUDA GPU or CPU also works
	Memory	16 GB unified memory (fp16, attention slicing)
	Storage	About 5 GB for model weights and generated files
Software	Frameworks	FastAPI, Uvicorn, Jinja2
	Libraries	google-generativeai, diffusers, torch, transformers, accelerate, fpdf2, Pillow, python-dotenv, pytest, httpx
	Development Environment	Python 3.14 venv, VS Code / Claude Code, Git
Data	Data	No training data: pretrained Stable Diffusion v1.5 and Gemini; user-entered text